Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policy frameworks coordinated through the Sustainable Banking and Finance Network (SBFN) have been adopted across more than sixty emerging and developing economies since 2012, on the premise that supervisory guidelines directing banks to price environmental risk into lending will strengthen the link between firms' access to credit and their green investment behaviour. We test this premise using two independent World Bank Enterprise Survey samples. The primary sample harmonizes the Bank's full 2018–2020 Green Economy Module rollout across 41 economies (28,042 firms) on a rich seven-item green-practice-adoption outcome, matched to each economy's SBFN policy-adoption status and Worldwide Governance Indicators regulatory quality as of its survey year. Because SBFN status is fixed within country in a single-wave cross-section, a classical country-fixed-effects specification cannot identify it or its interactions; we show this explicitly, then re-estimate the relationship with a Bayesian hierarchical model (country-varying slopes) and a causal forest that impose no linear-interaction functional form. All three approaches agree: bank credit access is robustly associated with green practice adoption (average marginal effect of 13 percentage points in

the baseline specification), but neither SBFN policy adoption nor regulatory quality moderates it—both institutional variables instead shift the general level of green adoption directly. We then subject this finding to the sharpest test we can construct: a second, independently assembled sample of 162 further economies (89,797–100,115 firms depending on specification), spanning 2021–2026 and reaching, for the first time in this literature, into high-income economies such as Australia, Germany, and Japan, built from a World Bank standardized cross-country database not previously exploited for this purpose. On a narrower CO2-emissions-monitoring outcome and with country fixed effects the primary sample's single-wave design cannot support, credit access remains significant while both SBFN and regulatory-quality interactions remain statistically indistinguishable from zero—the identical qualitative pattern, on a different outcome, a different set of economies, and a materially sharper identification strategy. The causal forest's feature-importance decomposition attributes the bulk of genuine effect heterogeneity in the primary sample (75.6%) to firm size rather than to either institutional moderator (regulatory quality 19.9%; SBFN status 2.6%). One finance-instrument-specific check finds a significant negative credit $\times$ SBFN interaction when finance access is measured via overdraft rather than term credit, consistent with—though not proof of—SBFN-guided lending being channelled preferentially through the term instruments that actually finance long-horizon green capital expenditure. We interpret the combined evidence as showing that green banking policy's binding constraint operates through firm-level absorptive capacity rather than through the country-level regulatory capacity its own institutional design targets.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks, banking regulators, and financial-sector regulatory bodies across the emerging and transition world have signed on to a coordinated effort to green their financial systems, adopting national guidelines that direct commercial banks to price environmental and climate risk into lending decisions and to channel credit toward cleaner production. On paper, this represents one of the most ambitious diffusions of environmental financial regulation outside the OECD: economies as different in income, banking-sector depth, and political regime as Bangladesh, Morocco, Mongolia, and Indonesia now report, to the International Finance Corporation's Sustainable Banking and Finance Network (SBFN), formal sustainable-finance roadmaps, green-taxonomy pilots, or mandatory environmental-risk disclosure rules for lenders. Whether that adoption changes what actually happens on a factory floor—whether a firm that can get a bank loan is any more likely to install a wastewater treatment system, switch to a more efficient boiler, or measure its own emissions—is a separate and considerably harder question, and it is the one this paper answers.

The premise that formal adoption of a policy is not the same as its operation in practice is, of course, not new to institutional economics. A large literature on state capacity and policy credibility has long argued that the same law, regulation, or guideline can produce entirely different real-economy outcomes depending on the administrative and judicial infrastructure available to enforce it (La Porta et al., 1998; Acemoglu et al., 2001). Applied to green finance specifically, a parallel body of work has documented that green credit guidelines in China measurably reshape lending allocation and firm environmental performance (Li et al., 2023; Huang et al., 2023; Dai et al., 2025), that firms' green investment responds to the financing constraints they face (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025), and that the World Bank Enterprise Surveys' own Green Economy module is a workable lens on firm-level environmental behaviour across dozens of countries at once (Asif et al., 2025; Phan, 2026). What is missing is the connection between these two literatures: a direct test, at firm level and across a wide cross-section of the very economies the green-banking-policy movement targets, of whether the credit-to-green-investment channel that these guidelines are designed to activate is actually stronger where the guidelines exist—and whether that depends on the state capacity to make the guideline credible in the first place.

This paper closes that gap using firm-level data from two independently assembled samples of World Bank Enterprise Survey (WBES) waves. The primary sample covers 47 country-years fielded between 2018 and 2023 that carry the Bank's full Green Economy module—a battery of questions on emissions monitoring, energy audits, and the adoption of climate-friendly production measures—alongside the Surveys' standard access-to-finance module; a second, global sample extends this to 162 further country-years fielded between 2021 and 2026, reaching for the first time in this literature into high-income economies with no SBFN involvement at all (Section 4). We match each economy to its Sustainable Banking and Finance Network policy status as of its survey year, distinguishing genuine ex-ante policy adopters from economies that joined only afterward, and to the World Bank's Worldwide Governance Indicators as a measure of the regulatory state capacity available to make that policy credible. Econometrically, we depart deliberately from the pooled fixed-effects specification that has become the default in this literature. Because SBFN status is fixed within country in a single-wave cross-section, a country fixed effect absorbs it entirely, and a naive interaction term estimated by pooled logit—which we report first, as a baseline, precisely to make this limitation visible rather than to hide it—returns no detectable policy-moderation effect at all. We show that this null result is not evidence of an absent relationship so much as a symptom of forcing a fundamentally hierarchical question (does a level-2, country-level policy variable reshape a level-1, firm-level financing relationship?) into a flat, single-level model with too few country clusters to lean on for inference (Cameron and Miller, 2015). We then re-estimate the same relationship with tools built for exactly this structure: a Bayesian hierarchical logistic model with country-varying slopes (Gelman and Hill, 2007; Gelman et al., 2020), a causal-forest / double-machine-learning (DML) estimator (Wager and Athey, 2018; Chernozhukov et al., 2018; Athey et al., 2019; Athey and Imbens, 2019) that lets the treatment effect of credit access vary flexibly across the institutional landscape rather than through a single estimated coefficient, and, on the much larger global sample, a country-fixed-effects specification that the primary sample's single wave cannot support.

This paper contributes to the literature in four respects. First, it provides what is, to our knowledge, the first direct empirical test of whether SBFN-style green banking policy adoption changes the firm-level relationship between finance and green investment, rather than only the aggregate volume of green lending reported by banks themselves—a firm-level, demand-side complement to the bank-level, supply-side evidence that dominates the existing green-credit-policy literature. Second, it extends the World Bank Enterprise Survey's Green Economy module—used until now mostly within single countries or single regional waves—into a harmonized 41-country primary sample spanning Europe, Central Asia, the Caucasus, and the Middle East and North Africa, giving the paper a breadth of institutional variation that a single-country or single-region design cannot offer. Third, and unusually for this literature, we subject the primary sample's central finding to an independently assembled second test: a 162-economy global sample built from a World Bank standardized cross-country database spanning 2021–2026 that, to our knowledge, no published study of green banking policy has yet exploited, extending coverage for the first time into high-income economies entirely outside the SBFN's own remit and allowing a country-fixed-effects specification the primary sample's single-wave design cannot support. That the same qualitative result—a robust finance-green channel, no detectable institutional moderation—survives this independent replication is, we think, considerably more informative than either sample could be alone. Fourth, methodologically, the paper demonstrates—with the same underlying question tested twice, on two different samples, under four different estimators—exactly where and why a classical fixed-effects approach fails to detect (or, when country fixed effects are usable, correctly fails to find) an institutionally-conditioned effect, offering applied researchers working with similarly structured cross-country firm data a concrete illustration of when the extra modelling investment is worth making and when a null result is simply a null result.

The paper also speaks directly to a growing body of recent scholarship in this journal on the institutional and financial determinants of firm and bank behaviour in emerging and transition economies. Yan et al. (2024) study how environmental protection taxes reshape green productivity among Chinese listed firms; Barra and Falcone (2026) examine how institutional factors condition environmental performance across a global sample of economies; Heyert and Weill (2025) and Bach et al. (2025) use firm- and bank-level data across dozens of countries to study, respectively, trust-driven financial inclusion and credit constraints' effect on corporate tax behaviour in Vietnamese firms; Li et al. (2026) study how shadow-banking engagement reshapes trade-credit financing among Chinese non-financial firms; and Horvath et al. (2025) apply Bayesian model averaging to the cross-country determinants of financial development, a methodological precedent we build on directly in adopting a Bayesian estimation strategy in Section 4. This paper sits at the intersection of these threads—institutions, green performance, and firm-level financing behaviour, studied jointly rather than pairwise—and, to our knowledge, is the first among them to test whether a specific, dateable policy-adoption event (SBFN membership) moderates the finance-green relationship at firm level across a comparably broad cross-section of adopting economies.

The remainder of the paper is organized as follows. Section 2 sets out the institutional background of the SBFN framework and the concepts and trends in green banking policy adoption across our sample. Section 3 develops the paper's three hypotheses from the state-capacity and institutional-complementarity literature. Section 4 describes the primary and global samples, variable construction, and the empirical strategy for each. Section 5 presents baseline, hierarchical, and heterogeneous-effect results on the primary sample, followed by the independent global-sample replication. Section 6 discusses the findings' economic magnitude and their implications for how green credit policy is designed and sequenced, and Section 7 concludes.

2. Institutional background: green banking policy in emerging and transition economies

The Sustainable Banking and Finance Network (SBFN)—established in 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an expanded mandate covering both banking and non-bank financial regulation—is the principal vehicle through which financial-sector regulators and banking associations in emerging and developing economies have coordinated the design of national sustainable-finance policy. Hosted and technically supported by the International Finance Corporation, the network now counts roughly seventy member institutions (central banks, banking regulators, and banking or securities associations) drawn overwhelmingly from Sub-Saharan Africa, East Asia and the Pacific, Europe and Central Asia, Latin America and the Caribbean, the Middle East and North Africa, and South Asia (International Finance Corporation / Sustainable Banking and Finance Network, 2025). Notably for our sample, the network's membership is essentially disjoint from the European Union: not one of the sixteen EU member states in our WBES sample appears on any SBFN membership roster we could locate, consistent with the network's explicit mandate to serve markets outside the reach of the EU's own sustainable-finance architecture (the Sustainable Finance Disclosure Regulation and the EU Taxonomy, both of which post-date most of our survey waves in any case).

Membership itself, however, is only the network's coarsest measure of policy adoption. SBFN's own Multi-Tiered Progress Framework—the instrument used in its periodic Global Progress Reports (2018, 2019, 2021, 2023–2024) and country-level Progress Report addenda—scores each member jurisdiction's actual regulatory output against several policy pillars, most centrally environmental-and-social risk management and, in later report vintages, ESG disclosure integration and green-finance-flow measurement, and places each jurisdiction on a progression from an initial "Commitment" or "Formulating" stage, through "Developing" or "Implementation," to "Advancing" and, for the most mature frameworks, "Consolidating." The variation this produces across our sample is considerable and is itself part of the paper's identifying variation, not just a robustness afterthought. Mongolia's Bank of Mongolia was a founding SBFN member in 2012 and had already reached the network's "Advancing" implementation stage by the time of the country's 2019 Enterprise Survey; Morocco's Bank Al-Maghrib, a member since 2014, sat at the same stage two years later. Bangladesh Bank, also a founding 2012 member, had progressed by its 2022 Enterprise Survey wave into the network's ESG-integration pillar at an "Advancing" classification, while India's Reserve Bank/Indian Banks' Association, a later (2016) entrant, remained at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan and the Kyrgyz Republic, both members since 2016 and 2018 respectively, had progressed only as far as an initial “Commitment” stage by their 2019 survey waves. A further, and for identification purposes equally important, group of economies in our sample joined SBFN only after their Enterprise Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajikistan (2023), and most of the Western Balkan and South Caucasus economies in our sample (2021–2024)—and are accordingly coded as non-adopters as of their survey year, even though several are now members. This ex-ante coding is deliberate: it is the policy environment a firm actually operated under at the moment its Green Economy module responses were recorded that is relevant to our test, not the policy environment that jurisdiction eventually adopted.

The economic logic behind the SBFN framework, and the reason it is a plausible channel for firm-level green investment rather than a purely symbolic commitment, rests on a straightforward supervisory mechanism: once a banking regulator formally adopts environmental-and-social risk management guidelines, commercial banks under its supervision face a compliance incentive to price environmental risk into individual lending decisions and, in many of the more advanced frameworks, to report a green-lending share to the supervisor directly. Existing evidence on China’s much-studied green credit guidelines documents exactly this channel operating at bank and firm level—reduced lending to high-polluting borrowers and improved environmental performance among firms that retain access to credit (Li et al., 2023; Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller literature on Bangladesh's own long-running green-banking guidelines points in the same direction (Khairunnessa et al., 2021). What neither literature has established is whether this channel is detectable across a broad, heterogeneous cross-section of SBFN-style adopters at the firm level using a common survey instrument, nor whether the strength of the channel depends on the regulatory capacity available to enforce the underlying guideline—the two questions this paper turns to next.

3. Literature review and hypothesis development

3.1. Access to finance and firm-level green investment

A firm's decision to install pollution-abatement equipment, retrofit machinery for energy efficiency, or absorb the upfront cost of an emissions-monitoring system is, first and foremost, an investment decision, and investment decisions are shaped by whether a firm can finance them. The theoretical case is close to a standard credit-constraint story: green investments of the kind captured in the WBES Green Economy module (heating and cooling upgrades, on-site renewable generation, waste-minimization retrofits, energy-management systems) typically carry a large upfront capital outlay against a payback stream realized only over several years, which is exactly the profile of investment most sensitive to a firm's access to external finance rather than to retained earnings alone. Firm-level evidence bears this out directly. Ghosh and Dutta (2022), using a panel of Indian manufacturing firms, show that credit-constrained firms are measurably less likely to adopt cleaner production practices even when they report an equivalent willingness to comply with environmental regulation. De Haas et al. (2025)

document, across a large cross-country firm sample, that managerial and financial barriers operate as distinct but jointly binding constraints on firms' green transition investments, and Kalantzis et al. (2022) reach a closely related conclusion using EBRD transition-economy survey data, finding that firms' green investment is driven by a mix of financing conditions and climate exposure rather than by climate motivation alone—precisely the joint finance-and-institutions framing this paper tests directly. Costa et al. (2024) and Truong (2026) find, respectively in OECD and Vietnamese data, that financing constraints blunt firms' green investment response to environmental policy pressure that would otherwise induce it, while Aristei and Gallo (2023) show that firms with better access to credit are both more likely to have adopted green management practices before a shock and better able to sustain them through one. Ullah (2025) narrows this further to unlisted SMEs specifically, showing that financial constraints bind corporate environmental responsibility at smaller firm scales—a firm-size channel this paper's own causal-forest evidence (Section 5) also implicates, though, as we report there, without recovering a clean monotonic size gradient of its own. Siewers et al. (2024) extend the firm-level environmental-performance literature to a global-value-chain setting, showing that a firm's position in international production networks shapes its environmental performance net of domestic financing conditions, and Bambe et al. (2026) show that environmental policy stringency itself reshapes firm efficiency in developing economies, underscoring that the firm-level margin, not only the aggregate or bank-level one, is where much of this literature's true test now lies. Taken together, this literature converges on a simple first-order proposition, which any test of a green-credit-policy channel must first establish before asking whether policy moderates it:

> **H1.** Firms with access to formal bank credit (a credit line or loan from a financial institution) are more likely to have adopted green production practices than otherwise-similar firms without such access.

3.2. Does green banking policy strengthen the finance-to-green-investment channel?

H1 describes a channel, not a policy. The entire premise of the SBFN framework and its national analogues—and of the much larger literature on green credit guidelines in China specifically (Li et al., 2023, 2024; Huang et al., 2023)—is that a supervisory mandate directing banks to price environmental risk into lending, or to preferentially finance green activity, should widen the gap in green-investment behaviour between firms with and without bank access relative to a counterfactual world without the policy: banks in adopting jurisdictions have a compliance incentive to seek out, and price favourably, exactly the borrowers undertaking the kind of investment our outcome variable measures, while credit-constrained firms in the same jurisdiction see no comparable improvement in their financing terms. This is a moderation hypothesis, not a simple main-effect one, and it is worth being explicit that most of the existing supporting evidence—again, overwhelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—is drawn from single-country studies of a single, unusually well-resourced green-credit regime. Whether the same interaction holds across the far more heterogeneous set of jurisdictions that have adopted an SBFN-style framework, many of them at a far earlier and less resourced stage of implementation than China's, is untested. This motivates:

> **H2.** The positive association between bank credit access and green-practice adoption (H1) is stronger in economies that had, as of the survey year, adopted an SBFN sustainable-finance policy framework than in economies that had not.

3.3. Institutional complementarity: does policy credibility depend on state capacity?

A green banking guideline is only as effective as the supervisory apparatus behind it. This is the institutional-complementarity argument at the centre of the comparative-institutions, law-and-finance, and state-capacity literatures. Hall and Soskice (2001)'s varieties-of-capitalism framework is the foundational statement of the underlying logic in its most general form: formally similar policy instruments produce systematically different outcomes depending on how they interlock with the surrounding institutional architecture—a regulatory mandate that functions as intended within one configuration of complementary institutions may function quite differently, or not at all, transplanted into another. Applied more narrowly to financial regulation specifically, the same logic runs through the law-and-finance and state-capacity literatures: La Porta et al. (1998) and the subsequent finance-and-growth literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al., 2005, 2008) established that formally identical financial-contracting rules produce very different real-economy financing outcomes depending on the quality of the legal and regulatory institutions that enforce them, while Acemoglu et al. (2001) showed the same logic operating at the level of a country's entire institutional endowment. Applied to our setting, the argument is direct: a bank supervisor's environmental-risk guideline can only reshape lending behaviour to the extent that the supervisor has the regulatory capacity—examination authority, credible sanctions, reliable disclosure enforcement—to make compliance with it more than nominal. Where that capacity is weak, an SBFN framework may exist on paper (a formal "Commitment" or "Formulating" stage classification, in the network's own terminology) without meaningfully altering bank lending behaviour on the ground, precisely the state-capacity/policy-credibility distinction that motivates this special issue. This is also the logic behind the small but growing strand of work on institutions and environmental performance published in this journal specifically: Barra and Falcone (2026), studying a global sample of economies, find that institutional-quality indicators condition environmental performance in ways that are not reducible to income level alone, a result our design is built to test at the firm rather than the country level. We proxy regulatory capacity with the World Bank's Worldwide Governance Indicators Regulatory Quality estimate, matched to each economy's survey year, and hypothesize a three-way interaction:

> **H3.** The policy-moderation effect described in H2 is itself increasing in regulatory quality: SBFN adoption strengthens the credit-to-green-investment channel most where the supervising jurisdiction has the institutional capacity to make the underlying guideline credible, and least—potentially not at all—where it

does not.

Hypotheses H2 and H3 are, by construction, statements about how a country-level variable reshapes a firm-level relationship—precisely the two-level structure that Section 4 argues a classical single-level fixed-effects regression is poorly suited to recover with a country-cluster count in the low forties, and that motivates the hierarchical and machine-learning estimators we turn to after reporting that classical benchmark.

4. Data and methodology

4.1. Data sources and sample construction

The paper's firm-level data come from the World Bank Enterprise Surveys (WBES), a standardized, stratified-random-sample survey of formal-sector, non-agricultural, non-extractive private firms with at least five employees, conducted by the World Bank's Enterprise Analysis Unit. Our primary analysis sample is the World Bank's own harmonized cross-country file combining the 41 Europe and Central Asia (ECA) and Middle East and North Africa (MENA) economies surveyed under a single, identical questionnaire between 2018 and 2020, the survey wave in which the Bank's Green Economy module—a battery of questions on emissions monitoring, energy-consumption targets, and the adoption of specific climate-friendly production measures—was fielded in full to every respondent. This yields 28,042 firm-level observations across 41 economies (Table 2). We supplement this primary sample with 15,556 additional firm observations from six further economies (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023, Timor-Leste 2021) in which a shortened, partially randomized version of the

Green Economy module was fielded; because the exact item wording and randomized sub-module assignment differ across these six economies, we do not pool them into the main harmonized outcome variable, and instead use them as an out-of-region external-validity check on a narrower, honestly-matched pair of indicators (CO2-emissions monitoring, available in five of the six; waste-minimization adoption, available in five of the six, with different specific economies covered by each), reported separately in Section 5.

The primary sample, while broad by the standards of the existing green-credit-policy literature, is not the full universe of WBES economies carrying *any* Green Economy content, and we build a second, independent sample to exploit the rest of it. The World Bank also maintains a standardized cross-country database (2006–2026) that harmonizes a common set of core variables across every Formal Sector Enterprise Survey wave it has ever fielded, 292,314 firm-level observations across 471 country-years. Within it, 160 country-years fielded between 2021 and 2026 — reaching, notably, into high-income economies entirely outside the SBFN's remit such as Australia, Canada, Germany, and Japan — carry a harmonized minimal Green Economy item set: climate-damage exposure and, more relevantly for our outcome construct, CO2-emissions monitoring, alongside the same standard access-to-finance module (credit line/loan, overdraft, perceived finance obstacle) used in the primary sample. We supplement this with Bangladesh 2022 and Indonesia 2023 — two economies whose Green Economy responses used an alternative randomized-module variable naming convention and so are absent from the standardized database's harmonized columns, but which we can recover from our own extraction of the underlying country files —

for a combined global sample of 162 economies and 100,115 firms (89,797 with complete data on all covariates). Firm size, sector, and productivity controls for this sample are drawn from the Bank's companion Productivity Database, merged by firm identifier. We refer to this as the *global sample*, and to CO2-emissions monitoring as its outcome variable throughout; it is the paper's vehicle for testing whether the primary sample's central finding replicates independently, at far greater scale, and under an identification strategy — country fixed effects — that the primary sample's single-wave design cannot support (Section 4).

Country-level data are drawn from two sources for both samples. For the primary sample's 47 economy-years, SBFN policy-adoption status, join year, and progression-framework stage as of each economy's survey year are compiled from the Sustainable Banking and Finance Network's Global Progress Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report addenda, and the network's live country-profile data portal, cross-checked against IFC press releases announcing individual member accessions; economies that joined SBFN only after their WBES survey year are coded as non-adopters as of that year (Section 2). For the global sample's remaining 157 economy-years, we draw membership status and join year directly from the SBFN data portal's full published member roster (67 members as of the portal's most recent update), applying the identical ex-ante rule: an economy is coded as an adopter only if its join year precedes its survey year. Six economy-years in the global sample (Central African Republic, Cameroon, Chad, Republic of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on the roster under a regional entry, "Central African States," which we interpret as referring to the CEMAC zone's common banking supervisor; four small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St. Vincent and the Grenadines, all 2025) are similarly recorded under "Eastern Caribbean States," which we interpret as the Eastern Caribbean Central Bank's member states. Both interpretations are flagged explicitly here as inferences rather than explicit per-country listings. Regulatory capacity is measured with the World Bank's Worldwide Governance Indicators (WGI) Regulatory Quality estimate, matched to each economy's exact survey year (or, where not yet published, the most recent prior year available) via the World Bank DataBank API; WGI does not cover Taiwan, which is accordingly dropped from global-sample regressions.

4.2. Variable construction

Outcome. Our primary dependent variable, *green practice adoption*, is a binary indicator equal to one if a firm reports having adopted at least one of seven Green Economy module practices over the preceding three years: more climate-friendly on-site energy generation, an energy-management system, waste-minimization or recycling measures, air-pollution control measures, other pollution-control measures, any energy-efficiency measure, or the use of on-site renewable energy. We also construct a continuous *green adoption index*, the mean of the same seven binary items, for robustness.

Treatment. *Bank credit access* is a binary indicator equal to one if the firm reports holding a line of credit or loan from a formal financial institution at the time of the survey.

Country-level moderators. *SBFN member* is a binary indicator equal to one if the economy had adopted a national SBFN sustainable-finance policy framework as of the survey year. *Regulatory quality* is the WGI Regulatory Quality estimate for the survey year (continuous, standardized to mean zero, unit standard deviation within the estimation sample for the hierarchical and machine-learning specifications).

Controls. Firm size (small/medium/large, per the Enterprise Surveys' own sampling strata), broad sector (manufacturing, services, construction), log sales, an exporter dummy (any direct or indirect export share), and a foreign-ownership dummy (any private foreign ownership share).

Table 1 lists every variable used in the analysis together with its exact WBES question code and source, in the interest of full transparency about variable construction given the multi-source nature of the merge.

4.3. Empirical strategy

We estimate the relationship between bank credit access and green practice adoption, and its moderation by SBFN status and regulatory quality, in four stages of increasing methodological sophistication, each addressing a specific limitation of the one before it.

Stage 1 — classical benchmark. We first estimate a pooled logistic regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\Big(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c}\boldsymbol{\gamma} + \delta_{s(i)}\Big) \tag{1}$$

for firm i in country c, where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a sector fixed effect, with standard errors clustered by country. We deliberately do not include country fixed effects in this specification: because SBFN_c is constant within country in a single cross-sectional wave, a country fixed effect would absorb it—and its interaction with $\text{Credit}_{i,c}$—completely, making β_2 and β_3 unidentifiable by construction. This is not a modelling choice we are free to make differently; it is the reason a purely classical fixed-effects approach cannot answer H2/H3 as posed, and we report this specification, extended with a triple interaction for Regulatory quality_c, precisely to make that limitation visible.

Stage 2 — Bayesian hierarchical (multilevel) model. We re-estimate the same relationship allowing country to enter as a random rather than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \tag{2}$$

$$\begin{aligned}\text{logit}(p_{i,c}) = (\beta_1 + u_{1,c})\,\text{Credit}_{i,c} + \beta_2\text{SBFN}_c + \beta_3(\text{Credit}_{i,c} \times \text{SBFN}_c)\\ + \beta_4(\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c}\boldsymbol{\gamma} + (\alpha + u_{0,c})\end{aligned} \tag{3}$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}\right) \tag{4}$$

estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS) in PyMC. This specification is the paper's primary vehicle for testing H2 and H3: because country enters through a variance component rather than a full set of dummies, the country-level moderators SBFN_c and RegQuality_c remain identified, while the random slope $u_{1,c}$ still absorbs unobserved country heterogeneity in the credit-green relationship and, together with partial pooling across countries, gives more defensible inference than cluster-robust standard errors computed over only 41 clusters (Cameron and Miller, 2015).

Stage 3 — causal forest / double machine learning. To avoid imposing a linear interaction functional form on how the treatment effect of credit access varies across the institutional landscape, we additionally estimate a Causal Forest with Double Machine Learning (Wager and Athey, 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted nuisance models for the outcome and treatment propensity, and firm size, export status, SBFN status, and regulatory quality as effect-modifying features. This lets us recover conditional average treatment effects (CATEs)—the estimated effect of credit access on green adoption, separately for SBFN members versus non-members, and separately by regulatory-quality tercile—without imposing that the moderation take the single-coefficient interaction form assumed in Stages 1–2.

Stage 4 — small-sample supplementary check. We additionally re-estimate a reduced-form version of the credit-green relationship using a waste-minimization-adoption outcome, available in four economies (India 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy modules asked this item under directly comparable wording. This outcome is not part of the standardized cross-country database underlying Stage 5 below, so it offers genuinely independent, if small-sample, supplementary evidence.

Stage 5 — global-sample replication with country fixed effects. Finally, and centrally, we re-estimate the full specification on the 162-economy global sample described in Section 4, using CO2-emissions monitoring as the outcome. Because this sample spans far more country clusters than the primary sample, we can now afford a country-fixed-effects specification: δ_c absorbs SBFN_c and RegQuality_c's main effects (collinear with the fixed effect by construction, since both are constant within country-year) while leaving their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access itself varies within country. This is a materially sharper test of H2/H3 than Stage 1's clustered-SE approach, on an entirely independent sample and outcome from Stages 1–3; we report it alongside a no-fixed-effects specification for direct comparability with the primary sample's Table 4.

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
`green_adoption_binary`	=1 if firm adopted ≥1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
`fin_has_credit_line`	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
`sbfn_member`	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
`wgi_regulatory_quality`	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (`GOV_WGI_RQ_EST`)
`size_cat`	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
`sector_broad`	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
`log_sales`	Log of total annual sales	D2	WBES General Information Module
`foreign_owned_dummy`	=1 if any foreign ownership share >0%	B2B	WBES Ownership Module
`exporter_dummy`	=1 if any export sales share >0%	D3B, D3C	WBES Sales Module

5. Results

5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herzegovina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

Table 2 reports the full 41-economy sample composition; firm counts range from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-practice-adoption rate ranges more than three-fold across economies, from 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside SBFN policy status: 8 of the 41 primary-sample economies had adopted an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyrgyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-economy sample once the five (of six) SBFN-member small-sample supplementary economies are counted—and every European Union member state in the sample is, consistent with SBFN's mandate, a non-adopter. At the firm level, bank credit access correlates positively with the green-adoption index ($r = 0.18$) but is essentially uncorrelated with a firm's own perceived finance obstacle ($r = -0.02$)—a first hint, well before any regression, that realized access to credit rather than perceived financing difficulty is what tracks green behaviour, and the two are evidently not interchangeable measures of the same underlying constraint.

Figure 2 previews the paper's central finding graphically before a single regression is estimated: plotting each economy's green-adoption rate against its regulatory-quality score and fitting separate lines for SBFN members and non-members produces two lines of very similar, modestly positive slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-member economies show systematically *lower* raw green-adoption rates at every level of regulatory quality, not a steeper relationship between regulatory quality and adoption. That combination—a level difference with no slope difference—is exactly the signature the regression evidence below confirms formally.

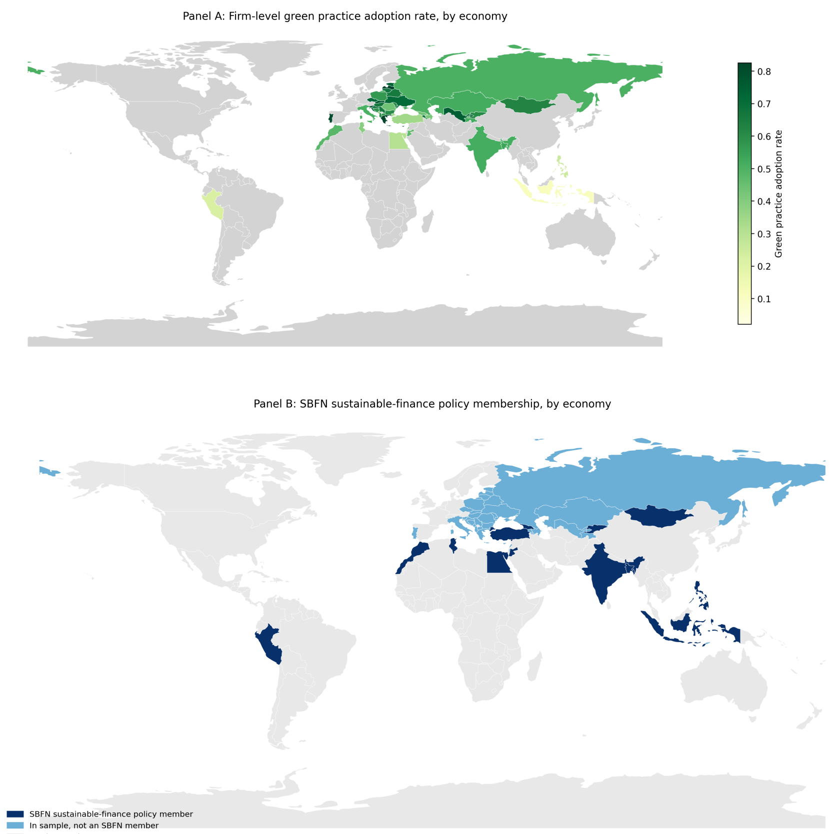

Figure 1: Green practice adoption and SBFN policy membership, by economy.

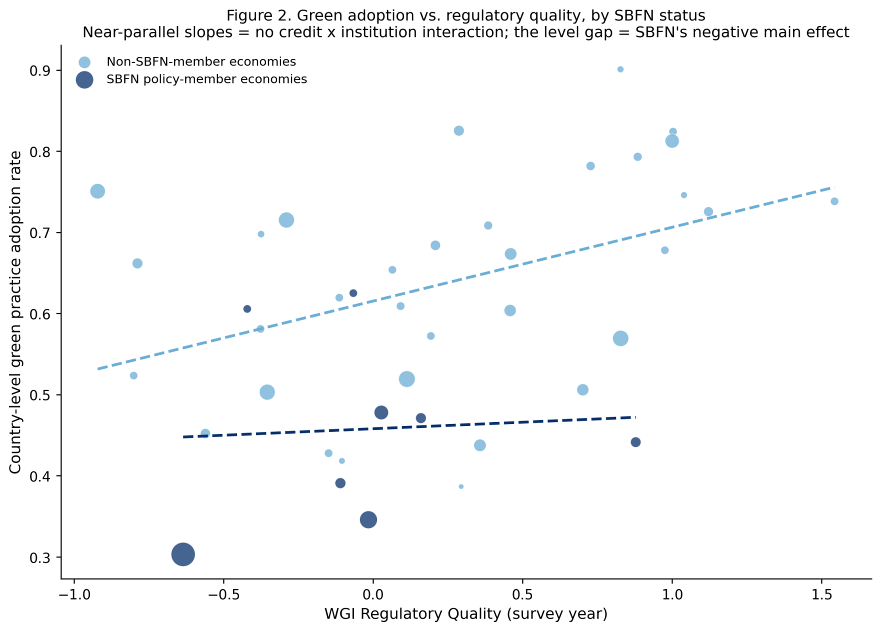

Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect $= 12.95$ percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3's fully saturated triple interaction with regulatory quality tells the same story at greater length: every interaction term involving credit access is small and insignificant, while regulatory quality's uninteracted relationship with adoption is positive, if only marginally significant in this particular fully-saturated specification ($b = 0.278$, $p = 0.110$).

As flagged in Section 4, this specification cannot include country fixed effects without mechanically absorbing the very SBFN and regulatory-quality terms the hypotheses concern, so what Table 3 shows is a genuine test of H2/H3, not a placeholder—and that test returns no interaction. The question the remaining two stages are built to answer is whether that null result reflects a real absence of policy moderation, or an artefact of forcing a two-level question through a single-level model with country-clustered standard errors computed over only 41 clusters, a setting in which such standard errors are known to be unreliable (Cameron and Miller, 2015).

5.3. Bayesian hierarchical model — primary test of H2/H3

The hierarchical specification—country-varying random slopes on credit access, cross-level interactions with SBFN status and standardized regulatory quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and ess_{bulk} between 797 and 2,270—is built specifically to give H2 and H3 a fair hearing that the absorbed classical specification could not. It does not change the conclusion. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval [0.18, 0.49]), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN mem-

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571***	0.459***	0.373***
	(0.123)	(0.090)	(0.085)
SBFN member		-0.991***	-0.844***
		(0.178)	(0.209)
Credit $\times$ SBFN		0.049	0.019
		(0.168)	(0.140)
Regulatory quality			0.278
			(0.174)
Credit $\times$ Reg. quality			0.144
			(0.142)
SBFN $\times$ Reg. quality			0.053
			(0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104
			(0.272)
Log sales	0.061	0.041	0.068**
	(0.039)	(0.034)	(0.032)
Foreign owned	0.219	0.244*	0.220
	(0.143)	(0.139)	(0.143)
Exporter	0.537***	0.471***	0.412***
	(0.092)	(0.074)	(0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

bership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2 and H3, neither cross-level interaction is distinguishable from zero: credit × SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit × regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we can state that SBFN status and regulatory quality shift the *level* of green adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit × SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit × Reg. quality	0.004	0.087	-0.13	0.14	1.00

5.4. Causal forest / heterogeneous treatment effects

Table 5 and Figure 5 report the Causal Forest DML estimates, which impose no linear-interaction functional form and let the treatment effect vary flexibly across firm and country covariates. The overall average treatment effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consistent with, though smaller in magnitude and less precisely estimated than, the logit average marginal effect of 0.130 in Section 5.2, reflecting the more conservative variance properties of a doubly-robust, cross-fitted nonparametric estimator relative to a parametric logit on the same data. Critically, the conditional average treatment effects triangulate with Section 5.3 rather than contradicting it: the estimated effect is 0.072 for firms in non-SBFN economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 / 0.075 across the low/middle/high regulatory-quality terciles respectively—three different institutional cuts, three essentially flat effects, all visibly overlapping in Figure 5. Two entirely independent estimators, one parametric and hierarchical, one nonparametric and forest-based, agree on the absence of institutional moderation.

The forest's feature-importance decomposition adds a genuinely new result rather than only confirming the null: of the four candidate effect-modifiers, firm size (log sales) accounts for 75.6% of the model's explained heterogeneity, versus 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter status. Whatever heterogeneity exists in how strongly credit access predicts green adoption is, on this evidence, primarily a firm-level rather than a country-institutional phenomenon. Collapsing this into the Enterprise Surveys' own coarse size strata, however, does not reveal a clean monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms, and 0.057 for large firms (Table 6; all with wide, overlapping confidence intervals spanning zero), which if anything runs opposite to a simple "larger firms better absorb a green investment's fixed costs" story rather than confirming it. Read alongside the 75.6% importance share for the continuous size measure, we take this as evidence that the heterogeneity the forest is detecting is more granular or non-monotonic across the firm-size distribution than a three-category collapse can reveal, rather than evidence for any particular direction of a size gradient—a qualification we report explicitly rather than paper over with a tidier-sounding but unsupported claim. This redirects, rather than closes, the paper's institutional-moderation question, and we return to it in Section 6.

5.5. Small-sample supplementary check: waste-minimization adoption

Before turning to the global-sample replication that is this paper's second major test (Section 5.7), Table 7 reports a smaller supplementary check using waste-minimization adoption — an outcome not captured in the standardized cross-country database underlying the global sample, and so a genuinely independent (if small-sample) data point — available in four economies whose Green Economy modules asked this item under directly comparable wording: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of these four are themselves SBFN members at various progression stages, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in Sections 5.2–5.4 is a peculiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes: pooled across the four economies, having an overdraft facility is associated

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE | SBFN non-member	0.0723	-0.0300	0.1746
CATE | SBFN member	0.0758	-0.0591	0.2107
CATE | reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE | reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE | reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	**Importance**		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

with significantly higher odds of waste-minimization adoption ($b = 0.422$, $p < 0.001$, $N = 9{,}038$, 4 countries). The direction of the core finance-green relationship replicates on a different continent, a different half-decade, and a narrower outcome definition than the one anchoring Sections 5.2–5.4 — a preview, on four economies, of the much larger replication in Section 5.7.

Table 7: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

5.6. *Additional robustness*

Table 8 subjects the primary-sample finding to four further checks, all retaining sector fixed effects, firm controls, and country-clustered standard errors (no country fixed effects, for the reason given in Section 4). First, replacing the binary outcome with the continuous seven-item green-adoption index and estimating by OLS reproduces the same pattern to the decimal: credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistinguishable from zero ($b = 0.006$, $p = 0.739$).

Second, replacing bank credit access with two alternative finance measures produces one genuine point of nuance worth reporting rather than smoothing over. Using overdraft-facility access instead of a credit line/loan, the main overdraft effect remains positive and significant ($b = 0.274$, $p =$

0.011), the SBFN main effect remains negative and significant ($b = -0.905$, $p < 0.001$)—but the overdraft × SBFN interaction is now negative and significant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper that clears a conventional significance threshold, and it runs opposite to the direction H2 predicts. Using the reverse-coded perceived finance-obstacle measure in place of realized access, by contrast, both the main effect and its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interaction $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation noted in Section 5.1. We read the overdraft result as suggestive rather than conclusive given it is one significant coefficient among more than a dozen interaction tests across the paper, but it is at least directionally consistent with a substantive story worth flagging for future work: if SBFN-guided lending is channelled preferentially through the term loans and credit lines that finance long-horizon capital expenditure, rather than through short-term overdraft facilities, we might expect exactly this pattern—a null-to-positive interaction on credit lines and a negative one on overdrafts—though our data cannot directly test whether SBFN-guided credit is disproportionately term-structured.

Third, splitting the sample into manufacturing ($N = 13{,}340$) and services ($N = 9{,}538$) subsamples shows the credit-access main effect, the SBFN main effect, and the null credit × SBFN interaction all replicate within each sector separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b = -0.040$, $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b = 0.195$, $p = 0.426$)—the paper's central null is not an artefact of pooling two structurally different production technologies.

Table 8: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

5.7. Global-sample replication with country fixed effects

The preceding five subsections establish the paper's central finding on the primary sample. We now subject that finding to what we consider the most demanding test available to us: complete re-estimation on an independently assembled, 162-economy global sample (Section 4), using a different outcome variable (CO2-emissions monitoring rather than the seven-item composite), different survey years (2021–2026 rather than 2018–2020), and, for the first time in the paper, a country-fixed-effects specification that this much larger set of clusters can support. Figure 3 maps the combined reach of the two samples together: 166 distinct economies, spanning every region and, through the global sample specifically, extending for the first time in this literature into high-income economies with no SBFN involvement at all.

Table 9 reports two specifications. Column M1, without country fixed effects, is the direct global-sample analogue of Table 4's primary-sample baseline: credit access remains positive and significant ($b = 0.198$, $p = 0.039$, $N = 89{,}797$, 159 countries), while the SBFN main effect ($b = -0.171$, $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and every regulatory-quality interaction are statistically indistinguishable from zero. Column M2 imposes country fixed effects, absorbing the SBFN and regulatory-quality main effects by construction (they are collinear with δ_c, since both are constant within a given country-year) while leaving their firm-level interactions with credit access fully identified, because credit access itself varies within country. This is a sharper test than anything the primary sample's single 2018–2020 wave can support, and it returns the same answer with even more precision: credit access is strongly significant ($b = 0.249$, $p < 0.001$), while the credit × SBFN interaction ($b = 0.506$, $p = 0.205$) and the credit × regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain null.

We regard this as the paper's most important robustness result. It is not a re-analysis of the same data under a different label: it is a different outcome construct, a different set of economies (only a handful of which overlap with the primary sample), a different half-decade, and an econometric specification — country fixed effects — that is only available at all because this second sample has enough clusters to support it. That the qualitative conclusion is identical across both samples and across the full set of four estimators reported in this paper (classical logit, Bayesian hierarchical, causal forest, and now country-fixed-effects logit) is, we think, considerably stronger evidence for the paper's central claim than either sample could offer alone.

5.8. Results summary across all specifications

Sections 5.2–5.7 report ten tables spanning four estimators and two independent samples. Figure 4 consolidates every coefficient bearing on H1–H3 into a single view. Panel A shows the credit-access effect (H1): every point estimate is positive and every confidence interval excludes zero, in both sam-

Table 9: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198**	0.249***
	(0.096)	(0.057)
SBFN member	-0.171	— (absorbed)
	(0.184)	
Credit × SBFN	0.278	0.506
	(0.352)	(0.399)
Regulatory quality (std.)	0.155	— (absorbed)
	(0.096)	
Credit × Reg. quality	0.092	-0.044
	(0.089)	(0.047)
Log sales	0.127***	0.288***
	(0.027)	(0.028)
Foreign owned	0.621***	0.528***
	(0.083)	(0.046)
Exporter	0.456***	0.335***
	(0.094)	(0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

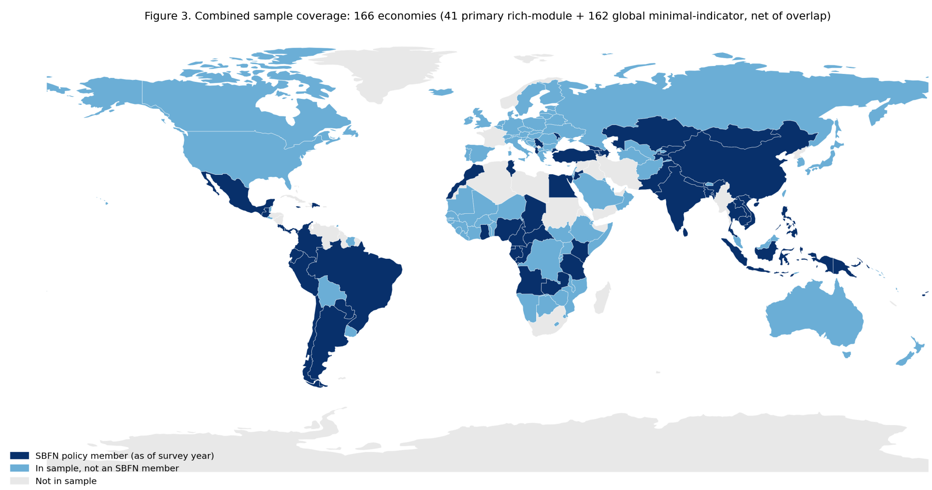

Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

ples and all four estimators. Panels B and C show the credit × SBFN and credit × regulatory-quality interactions (H2, H3): every single confidence interval, across both samples and every estimator, straddles zero. Figure 5 shows the same non-result from a completely different angle—the causal forest's conditional average treatment effects broken out by SBFN status, regulatory-quality tercile, and firm-size category all overlap both each other and the overall average treatment effect. No subgroup split, by any variable examined in this paper, recovers a distinguishable effect.

6. Discussion

Put in plain economic terms, the paper's central finding is this: a firm holding a bank credit line is roughly 13 percentage points more likely to have adopted a green production practice than an otherwise identical firm without

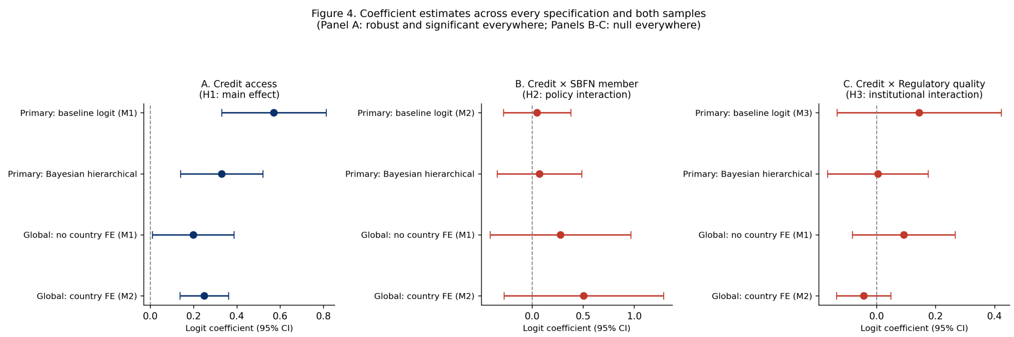

Figure 4: Coefficient estimates across every specification and both samples.

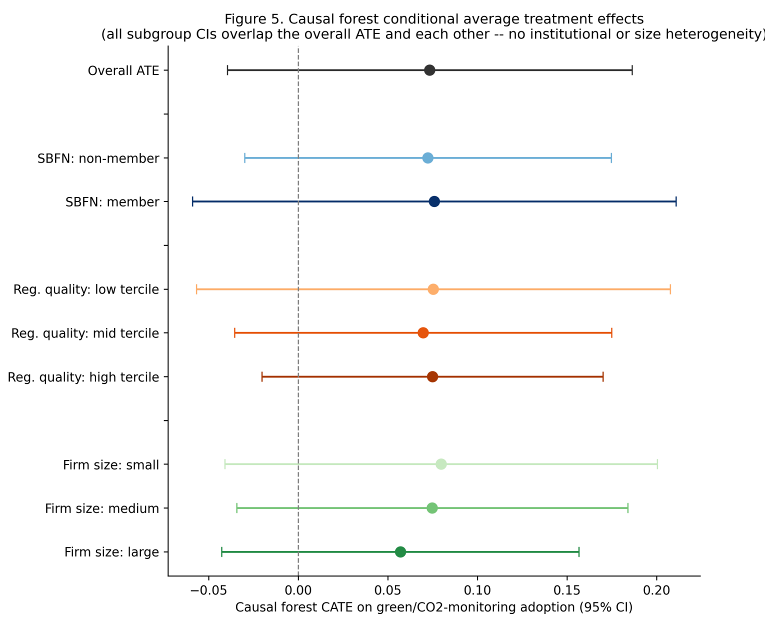

Figure 5: Causal forest conditional average treatment effects, by subgroup.

one—comparable in magnitude to the gap between a small firm and a large one—and that gap is essentially the same size whether the firm sits in an SBFN sustainable-finance policy adopter or a non-adopter, and essentially the same size whether the country's regulatory quality sits in the bottom, middle, or top third of the sample (Figure 4). Firms in SBFN-adopting economies are, on average, some 15–20 percentage points *less* likely to have adopted a green practice at all—a level gap, not a slope gap.

This convergence matters most because of how differently the two samples could have failed to agree. The primary and global samples share almost no economies in common, were fielded in different half-decades, measure green practice adoption with entirely different survey items, and reach institutional settings—high-income, non-SBFN economies such as Australia and Germany—that the primary sample cannot touch at all. A finding that held only in the 41-country ECA-MENA sample could plausibly be a regional or period-specific artefact; a finding that survives unchanged when re-estimated on 162 largely non-overlapping economies under a sharper identification strategy is considerably harder to explain away as such.

This is worth dwelling on precisely because it complicates rather than confirms the implicit premise of much of the existing green-credit-policy literature, which is disproportionately built on evidence from China's unusually mature, unusually well-resourced green credit guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025). Our sample includes economies at every stage of the SBFN's own progression framework—from Jordan and the Kyrgyz Republic's initial "Commitment" stage to Mongolia's "Advancing" implementation a full seven years after founding membership—and across that entire range, we find no evidence that formal policy adoption changes how a firm's credit access translates into green investment behaviour. Section 5.4's most novel result gives one plausible redirection of the question, though not a tidy answer to it: the heterogeneity that *does* exist in the credit-green relationship is overwhelmingly a firm-size phenomenon (75.6% of explained variance) rather than a country-institutional one (regulatory quality 19.9%, SBFN status a mere 2.6%)—yet, as Section 5.4 reports honestly, collapsing this into the Surveys' own coarse Small/Medium/Large strata does not recover a clean gradient in either direction, so we cannot yet say whether it is small or large firms for whom the credit-green channel runs stronger, only that firm scale—in some more granular or non-monotonic form than a three-category split reveals—is where the forest locates most of the effect heterogeneity a country-level policy variable does not. What this does support is a reorientation of where policy attention belongs: if the operative constraint sits at the level of which individual firms can absorb a green investment's fixed costs once financed, rather than at the level of whether a supervisory guideline nudges banks to lend more broadly, then a country-level supervisory mandate operating on the extensive margin of bank lending decisions may be aimed at the wrong margin regardless of its own institutional credibility—a question future work with a finer-grained firm-size or capacity measure is better placed to resolve than the coarse categories available here.

The one interaction that does clear a conventional significance threshold—a *negative* credit $\times$ SBFN effect when finance access is measured through overdraft facilities rather than term credit lines (Section 5)—is, we think, more consistent with this redirection than with a simple failure of green banking policy. Overdrafts finance working capital, not the kind of multi-year capital expenditure our outcome variable measures; if SBFN-guided lending is disproportionately channelled through the term-loan instruments better suited to financing a solar installation or an energy-efficiency retrofit, exactly the pattern we find—a null-to-positive effect on credit lines, a negative one on overdrafts—is what a supervisor successfully steering credit toward long-horizon green investment, rather than short-horizon working capital, would produce. We flag this as suggestive rather than established, since it is a single significant coefficient among the many interaction tests reported across Tables 3–8, and our data cannot directly verify the underlying loan-tenor mechanism.

Read against this special issue's organizing themes of state capacity and policy credibility, our results sit closer to a state-capacity-is-necessary-but-firm-level-frictions-bind-independently story than to a straightforward institutional-complementarity one in which stronger regulatory quality unlocks a latent policy effect that would otherwise lie dormant. Regulatory quality matters—it raises the general baseline propensity to adopt green practices regardless of a firm's own financing arrangement—but it does not appear to be the missing ingredient that would let a green banking guideline reshape the credit-to-green channel specifically. This suggests, tentatively, that the binding constraint on green credit policy's effectiveness in our sample is less about whether the supervising jurisdiction has the state capacity to make the guideline credible, and more about whether firms below a certain scale can absorb a green investment's fixed costs at all once credit is available—a question about firm-level absorptive capacity that sits one level down from the state-capacity question the SBFN framework itself is designed to address.

7. Conclusion

This paper set out to test whether the wave of green banking policy adoption coordinated through the Sustainable Banking and Finance Network changes the relationship between firm-level access to finance and green investment behaviour, and whether any such moderation depends on the regulatory capacity available to make the underlying policy credible. Using harmonized World Bank Enterprise Survey data spanning 41 economies' full 2018–2020 Green Economy Module rollout, a four-country waste-minimization supplementary check, and an independently assembled 162-economy global sample spanning 2021–2026 and reaching for the first time into high-income, non-SBFN economies, we apply a deliberately escalating sequence of estimators—a classical benchmark chosen to expose its own limitation, a Bayesian hierarchical model built to give the moderation hypothesis a fair test, a causal forest imposing no functional-form assumption at all, and, on the global sample specifically, a country-fixed-effects specification the primary sample's single wave cannot support—and find four consistent results. First, access to bank credit is robustly associated with a firm's green practice adoption, a relationship that survives every specification, subsample, and alternative-outcome check in the paper and replicates independently in both the small waste-minimization sample and the full 162-economy global sample. Second, neither SBFN policy adoption nor regulatory quality moderates that relationship in any of the four estimators, on either sample, though both shift the general level of green adoption directly and substantially in the primary sample. Third, where genuine effect heterogeneity does exist, it is concentrated at the level of firm size rather than country institutions. Fourth, and most importantly for the paper's overall credibility, this entire pattern replicates on a sample sharing almost no economies with the primary one, fielded years later, measured with a different outcome item, and identified under a materially sharper econometric strategy—the single strongest piece of evidence in the paper that the null on institutional moderation is a real feature of the data rather than an artefact of any one sample, period, or estimator.

For policy, the implication is not that green banking guidelines are ineffective, but that our evidence does not support treating them as an effective lever specifically for widening the gap in green investment between financed and unfinanced firms—the margin most green-credit-policy literature implicitly assumes they operate on. If our tentative loan-tenor interpretation of the overdraft-interaction finding is correct, policymakers designing or refining SBFN-style frameworks may find more traction focusing supervisory attention on the term-lending instruments that actually finance green capital expenditure, and on complementary measures—technical assistance, matching grants, or credit guarantees—that address smaller firms' absorptive-capacity constraints directly, rather than assuming that a bank-level supervisory mandate alone will reach firms regardless of scale.

We close with five limitations that bound how far these conclusions travel. First, both samples are single cross-sections per country (Section 2 discusses why a genuine panel on this outcome does not exist in the published WBES data): they cannot rule out reverse causality (firms already predisposed to green investment may find it easier to obtain credit) or unobserved, timeinvariant country characteristics correlated with both SBFN adoption and the baseline propensity to adopt green practices, and our identification rests on firm-level variation in credit access interacting with a plausibly firm-exogenous country-level policy indicator, not a natural experiment; the global sample's country fixed effects sharpen this only for the interaction terms, not for the credit main effect itself. Second, SBFN membership is a binary proxy for what the network's own multi-tiered progression framework treats as a genuinely graded process; a natural extension would model policy stage—Commitment through Consolidating—as an ordinal or continuous treatment rather than collapsing it to membership status, which our binary coding necessarily does, and which for the global sample's regional entries (the CEMAC and Eastern Caribbean groupings, Section 4) additionally rests on an inference about which specific economies a regional listing covers rather than an explicit per-country statement. Third, the four-economy waste-minimization check cannot re-test the SBFN contrast itself, since three of its four economies are SBFN members, and combines non-identical outcome items across countries by construction of the WBES's own shortened, randomized module design in later survey waves—a genuine data-availability constraint rather than a methodological choice. Fourth, the global sample's CO2-monitoring outcome is a single item rather than the primary sample's seven-item composite, and is a narrower construct than "green practice adoption" in the sense the primary sample measures it; that the same qualitative pattern holds on this narrower measure is reassuring but does not establish that it would hold on a richer one, were a richer one available at global scale. Fifth, the Worldwide Governance Indicators' regulatory-quality measure is a broad, economy-wide governance indicator rather than a banking-supervision-specific capacity measure; a supervisor-specific index, were one available consistently across this country set, would sharpen the institutional-complementarity test considerably. We leave each of these to future work.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Claude (Anthropic) to assist with data extraction and cleaning from World Bank Enterprise Surveys microdata (including identification and extraction of the standardized cross-country database underlying the global sample), construction of the analysis datasets and merge with country-level SBFN and Worldwide Governance Indicators data, execution of the statistical and machine-learning analyses reported in Section 5, generation of Figures 1–3, and drafting of portions of the manuscript text. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. SBFN policy status and regulatory quality, full sample

Table A.10 reports, for all 47 economy-years in the paper's primary and small-sample supplementary check, the SBFN membership status coded for the analysis, the join year where applicable, the SBFN Multi-Tiered Progress Framework stage in force nearest the survey year, and the exact year of the Worldwide Governance Indicators observation used. Full per-country source citations (the specific SBFN Global Progress Report, country addendum, or accession announcement consulted for every determination) are available in the online supplementary material. The analogous 162-economy table for the global sample (Section 4) is summarized by region in Appendix C and provided in full in the online supplementary material and the replication package.

Table A.10: SBFN policy status and regulatory quality, full 47-economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integration)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Preparation)	2020
Estonia 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Georgia 2019	1	2017	Developing (Implementation)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019
India 2022	1	2016	Developing (Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)	2023
Italy 2019	0	—	None	2019
Jordan 2019	1	2016	Commitment (Preparation)	2019
Kazakhstan 2019	0	—	None	2019
Kosovo 2019	0	—	None	2019
Kyrgyz Republic 2019	1	2018	Commitment (Preparation)	2019
Latvia 2019	0	—	None	2019
Lebanon 2019	0	—	None	2019
Lithuania 2019	0	—	None	2019
Malta 2019	0	—	None	2019
Moldova 2019	0	—	None	2019
Mongolia 2019	1	2012	Advancing (Implementation)	2019
Montenegro 2019	0	—	None	2019
Morocco 2019	1	2014	Advancing (Implementation)	2019
North Macedonia 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Peru 2023	1	2013	Advancing (ESG Integration)	2023
Philippines 2023	1	2013	Advancing (ESG Integration)	2023
Poland 2019	0	—	None	2019
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Republic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementation)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

Appendix B. Bayesian model convergence diagnostics

Convergence diagnostics for every focal parameter of the hierarchical model reported in Table 4 are included directly in that table ($\hat{R}$ column); the full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member) to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional minimum threshold of 400 effective samples per parameter for stable posterior summaries at four chains. No divergent transitions were flagged during sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup draws each after 1,000 tuning iterations).

Appendix C. Global-sample SBFN status, regional summary

Of the 162 global-sample economy-years, 61 are coded as SBFN members as of their survey year and 101 as non-members. Table C.11 reports the regional distribution of the 61 members, drawn from the SBFN data portal's regional classification; five of the 61 (Peru, Philippines, Timor-Leste, Bangladesh, and Indonesia) carry the deeper per-country sourcing already reported in Appendix A, having been determined during the primary sample's original research pass rather than from the portal roster alone.

Additional material

Online appendix

Appendix A, Appendix B, and Appendix C appear above; per-country source citations at the level of detail compiled during data construction (individual SBFN Global Progress Report page references, country-addendum

Table C.11: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61

URLs, and IFC accession-announcement links for every non-obvious determination in the primary sample; the full 162-row global-sample SBFN/WGI table underlying Appendix C) are available in the online supplementary material accompanying this submission.

Data availability

Data in support of the findings of this study is available from the corresponding author upon reasonable request, subject to the World Bank Enterprise Surveys' own terms of use for the underlying firm-level microdata (enterprisesurveys.org, free registration required), including the standardized cross-country database and Productivity Database underlying the global sample. The country-level SBFN policy-status (including the SBFN data portal's member roster underlying the global sample) and Worldwide Governance Indicators datasets constructed for this paper, together with the full analysis and replication code, are available in a public GitHub repository;

the repository link is withheld from this anonymized manuscript, and repository access is restricted for the duration of double-anonymized review, to avoid identifying the authors to reviewers. The link will be provided to the editor upon submission and made publicly accessible again to readers upon acceptance.

References

Acemoglu, D., Johnson, S., Robinson, J.A., 2001. The colonial origins of comparative development: An empirical investigation. American Economic Review 91, 1369–1401. doi:10.1257/aer.91.5.1369.

Aristei, D., Gallo, M., 2023. Green management, access to credit, and firms' vulnerability to the COVID-19 crisis. Small Business Economics , 1–33doi:10.1007/s11187-023-00759-1.

Asif, M., Dhamija, A., Khan, W., Johri, A., Wasiq, M., 2025. Factors affecting adoption of green practices by food firms: evidences from the World Bank enterprises survey. Clean Technologies and Environmental Policy 27, 4841–4858. doi:10.1007/s10098-024-03122-4.

Athey, S., Imbens, G.W., 2019. Machine learning methods that economists should know about. Annual Review of Economics 11, 685–725. doi:10.1146/annurev-economics-080217-053433.

Athey, S., Tibshirani, J., Wager, S., 2019. Generalized random forests. The Annals of Statistics 47, 1148–1178. doi:10.1214/18-AOS1709.

Bach, T.N., Le, T., Luong, B.T., Vu, H.V., 2025. Credit constraints and corporate tax payments: Evidence from Vietnamese firms. Economic Systems 49, 101305. doi:10.1016/j.ecosys.2025.101305.

Bambe, B.W.W., Combes, J.L., Kouassi, B.V., Schwartz, S., 2026. Environmental policy stringency and firm efficiency in developing countries. Small Business Economics 66, 1163–1182. doi:10.1007/s11187-025-01128-w.

Barra, C., Falcone, P.M., 2026. Institutional factors and environmental performance: Insights from global economies. Economic Systems 50, 101364. doi:10.1016/j.ecosys.2025.101364.

Beck, T., Demirgüç-Kunt, A., Maksimovic, V., 2005. Financial and legal constraints to growth: Does firm size matter? The Journal of Finance 60, 137–177. doi:10.1111/j.1540-6261.2005.00727.x.

Beck, T., Demirgüç-Kunt, A., Maksimovic, V., 2008. Financing patterns around the world: Are small firms different? Journal of Financial Economics 89, 467–487. doi:10.1016/j.jfineco.2007.10.005.

Cameron, A.C., Miller, D.L., 2015. A practitioner's guide to cluster-robust inference. Journal of Human Resources 50, 317–372. doi:10.3368/jhr.50.2.317.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J., 2018. Double/debiased machine learning for treatment and structural parameters. The Econometrics Journal 21, C1–C68. doi:10.1111/ectj.12097.

Costa, H., Demmou, L., Franco, G., Lamp, S., 2024. The role of financing constraints and environmental policy on green investment. Economics Letters 239, 111741. doi:10.1016/j.econlet.2024.111741.

Dai, Q., He, J., Guo, Z., Zheng, Y., Zhang, Y., 2025. Green finance for sustainable development: analyzing the effects of green credit on high-polluting firms' environmental performance. Humanities and Social Sciences Communications 12, 854. doi:10.1057/s41599-025-05218-8.

De Haas, R., Martin, R., Muûls, M., Schweiger, H., 2025. Managerial and financial barriers to the green transition. Management Science 71, 2890–2921. doi:10.1287/mnsc.2023.00772.

Demirgüç-Kunt, A., Maksimovic, V., 1998. Law, finance, and firm growth. The Journal of Finance 53, 2107–2137. doi:10.1111/0022-1082.00084.

Gao, D., Mo, X., Duan, K., Li, Y., 2022. Can green credit policy promote firms' green innovation? evidence from China. Sustainability 14, 3911. doi:10.3390/su14073911.

Gelman, A., Hill, J., 2007. Data Analysis Using Regression and Multilevel/Hierarchical Models. Cambridge University Press. doi:10.1017/CBO9780511790942.

Gelman, A., Hill, J., Vehtari, A., 2020. Regression and Other Stories. Cambridge University Press. doi:10.1017/9781139161879.

Ghosh, D., Dutta, M., 2022. Environmental behaviour under credit constraints – evidence from panel of Indian manufacturing firms. Structural

Change and Economic Dynamics 63, 490–500. doi:10.1016/j.strueco.2022.07.004.

Hall, P.A., Soskice, D., 2001. Varieties of Capitalism: The Institutional Foundations of Comparative Advantage. Oxford University Press.

Heyert, A., Weill, L., 2025. Trust in banks and financial inclusion: Micro-level evidence from 28 countries. Economic Systems 49, 101248. doi:10.1016/j.ecosys.2024.101248.

Horvath, R., Horvatova, E., Siranova, M., 2025. The determinants of financial development: Evidence from Bayesian model averaging. Economic Systems 49, 101274. doi:10.1016/j.ecosys.2024.101274.

Huang, Z., Gao, N., Jia, M., 2023. Green credit and its obstacles: Evidence from China's green credit guidelines. Journal of Corporate Finance 82, 102441. doi:10.1016/j.jcorpfin.2023.102441.

International Finance Corporation / Sustainable Banking and Finance Network, 2025. Sustainable banking and finance network global progress report 2025. World Bank Group / IFC. https://documents1.worldbank.org/curated/en/099634202132617625/pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf.

Kalantzis, F., Schweiger, H., Dominguez, S., 2022. Green Investment by Firms: Finance or Climate Driven? Technical Report. EBRD Working Paper No. 268. doi:10.2139/ssrn.4132545.

Khairunnessa, F., Vazquez-Brust, D.A., Yakovleva, N., 2021. A review of the recent developments of green banking in Bangladesh. Sustainability 13, 1904. doi:10.3390/su13041904.

La Porta, R., Lopez-de Silanes, F., Shleifer, A., Vishny, R., 1998. Law and finance. Journal of Political Economy 106, 1113–1155. doi:10.1086/250042.

Levine, R., 2005. Finance and growth: Theory and evidence, in: Aghion, P., Durlauf, S. (Eds.), Handbook of Economic Growth. Elsevier. volume 1A. chapter 12, pp. 865–934. doi:10.1016/S1574-0684(05)01012-9.

Li, C., Feng, X., Li, X., Zhou, Y., 2023. Effect of green credit policy on energy firms' growth: evidence from China. Economic Research-Ekonomska Istraživanja 36. doi:10.1080/1331677X.2023.2177701.

Li, C., Liu, Z., Song, R., Zhang, Y.J., 2024. The impact of green credit guidelines on environmental performance: Firm-level evidence from China. Technological Forecasting and Social Change 205, 123524. doi:10.1016/j.techfore.2024.123524.

Li, Y., Liu, X., Li, J., Gao, X., 2026. Shadow banking engagement and trade-credit financing: Evidence from nonfinancial firms in China. Economic Systems 50, 101351. doi:10.1016/j.ecosys.2025.101351.

Nguyen, M.P., Phan, A., 2025. Impact of green credit policy on the operation of Vietnamese commercial banks: An empirical study using difference-in-differences model. PLOS One 20, e0333759. doi:10.1371/journal.pone.0333759.

Phan, Q.T., 2026. The impact of green economy practices on firm innovation using data from the World Bank enterprise surveys. Discover Sustainability 7, 848. doi:10.1007/s43621-026-03226-5.

Siewers, S., Martínez-Zarzoso, I., Baghdadi, L., 2024. Global value chains and firms' environmental performance. World Development 173, 106395. doi:10.1016/j.worlddev.2023.106395.

Tran, T.T.C., Nguyen, L., Do, T.K., 2026. The impact of fintech credit on green innovation: international evidence. Journal of Economics and Development 28, 111–127. doi:10.1108/JED-05-2025-0177.

Truong, D.D., 2026. Does access to finance condition firms' green investment responses to environmental pressure? evidence from Vietnam. PLOS One 21, e0341960. doi:10.1371/journal.pone.0341960.

Ullah, B., 2025. Corporate environmental responsibility and financial constraints for unlisted SMEs. Global Finance Journal 67, 101176. doi:10.1016/j.gfj.2025.101176.

Wager, S., Athey, S., 2018. Estimation and inference of heterogeneous treatment effects using random forests. Journal of the American Statistical Association 113, 1228–1242. doi:10.1080/01621459.2017.1319839.

Yan, Z., Jia, Y., Zhang, B., 2024. Environmental protection taxes and green productivity: Evidence from listed companies in China. Economic Systems 48, 101213. doi:10.1016/j.ecosys.2024.101213.